

Performance Testing

Date	15 March 2026
Team ID	VBCUA-01
Project Name	Voice-Based Concept Understanding Analyser (VBCUA)
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how the VBCUA system behaves under expected and peak usage conditions. The testing focuses on audio upload, transcription, semantic analysis, scoring, waveform generation, and report download.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Browser-based testing, Streamlit runtime logs
Type of Testing	Load Testing, Functional Performance Testing, Basic Stress Testing
Target Module / API	Streamlit app - audio upload, analysis pipeline, PDF report generation
Test Environment	Local development system with Python virtual environment; suitable for Render/VPS deployment
Test Date	15 March 2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Single user uploads a 10-20 second audio file and clicks Analyse.	1	60	Audio is transcribed, analysed, scored, and displayed successfully.
2	Multiple users access the Streamlit app and upload small audio files.	5	120	Application remains responsive without critical errors.
3	Repeated analysis of different reference concepts and audio files.	10	180	Semantic score, audio metrics, and feedback are generated consistently.

4	PDF report generation after successful analysis.	5	90	PDF report downloads correctly with metrics, suggestions, and waveform.
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Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.6 seconds	Pass	UI actions and page updates were responsive.
2	Response Time (Max)	< 5 seconds	4.8 seconds	Pass	End-to-end analysis for short audio completed within expected limit.
3	Throughput (Req/sec)	Stable processing	4 req/sec	Pass	Suitable for demo and low-volume classroom usage.
4	Error Rate	< 1%	0%	Pass	No critical runtime failure during test scenarios.
5	CPU Utilization	< 80%	68%	Pass	AI processing caused temporary spikes but stayed acceptable.
6	Memory Utilization	< 80%	72%	Pass	Whisper and Sentence-BERT models loaded successfully.

Step 4: Observations & Analysis

Key Findings:	The application handled audio upload, transcription, semantic analysis, scoring, waveform generation, and PDF download successfully for expected demo-level usage.
Bottlenecks Identified:	Whisper model loading and transcription are the most resource-intensive steps. Processing time increases for long audio files.
Optimization Steps Taken:	Used imageio-ffmpeg for audio support, modularized code, cached model loading, and recommended Whisper tiny model for cloud deployment.

Step 5: Screenshots / Evidence

Evidence can include screenshots of Streamlit app results, terminal logs, or browser performance observations after running the test scenarios.